Markscheme

Mathematics: analysis and approaches
Practice question bank

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145. (a)

The locus $|z-4| = 2|z-1|$ represents a circle (an Apollonius circle).

Find the coordinates of the centre and the radius of this circle.

let $z = x + yi$: $(x-4)^2 + y^2 = 4\left[(x-1)^2 + y^2\right]$ *(M1)*

expand: $x^2 - 8x + 16 + y^2 = 4x^2 - 8x + 4 + 4y^2$ *(M1)* **A1**

simplify: $-3x^2 - 3y^2 + 12 = 0 \Rightarrow x^2 + y^2 = 4$ *(M1)*

centre $(0, 0)$, radius 2 **A1**

[6 marks]